System Rules

Task: Generate a data science pipeline in Python 3.10.

Input: A dataset in CSV format, a schema that describes the columns and data types of the dataset, and a data profiling info that summarizes the statistics and quality of the dataset.

Output: A pipeline code that performs the following steps:

#1: Import the necessary libraries and modules.

#2: Load the training and test datasets. For the training data, utilize the variable "train_data.csv", and for the test data, employ the variable "test_data.csv".

#3: The user will provide the Schema, and Data Profiling Info of the dataset with columns appropriately named as attributes.

#4: Perform missing value imputation for features 'Address' and 'Zip'.

#5: Perform feature extraction (dataset contains categorical values).

#6: Perform feature selection (redundant columns e.g., 'Address' and 'Zip').

#7: Select an appropriate ML algorithm.

#8: Assign a default value to a particular hyperparameter.

#9: Evaluate the model.

Catalog Data

Dataset Description: A The dataset was obtained from multiple sources, including surveys, job posting sites, and other publicly available sources.A total of 6704 data points were collected.The dataset included fivevariables: age, experience, job role, and education level and salary.

Schema, and Data Profiling Info:

```
# Experience (string), distinct [ 60% ], missing [ 0% ], categorical-vals [1 year, 2 years, 3 years]
# Gender (string), distinct [ 40% ], missing [ 0% ], categorical-vals [ Male, Female ]
# Address (string), distinct [ 40% ], missing [ 20% ], categorical-vals [ CA, TX ]
# Zip (string), distinct [ 40% ], missing [ 40% ], categorical-vals [ 7050, 7871 ]
# Salary (int, target feature), min-max vals [ 100, 310 ], total-vals [ 5 ]
```

Categorical Features: Experience, Gender, Address, Zip

```python
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score

train_data = pd.read_csv("train_data.csv")
test_data = pd.read_csv("test_data.csv")

categorical_features = ["Experience", "Gender", "Address", "Zip"]
categorical_transformer = Pipeline(steps=[
      ("imputer", SimpleImputer(strategy="most_frequent")),
      ("onehot", OneHotEncoder(handle_unknown="ignore"))])

preprocessor = ColumnTransformer(
      transformers = [("cat", categorical_transformer, categorical_features)],
      remainder="drop" )

model = RandomForestRegressor(n_estimators=100, random_state=42)
pipeline = Pipeline(steps=[ ("preprocessor", preprocessor), ("model", model)])

X_train = train_data.drop("Salary", axis=1)
y_train = train_data["Salary"]
X_test = test_data.drop("Salary", axis=1)
y_test = test_data["Salary"]

pipeline.fit(X_train, y_train)
y_test_pred = pipeline.predict(X_test)

test_r2 = r2_score(y_test, y_test_pred)
print(f"Test $R^2$: {test_r2}")
```